

Lab Procedure for Python

Forward Kinematics

Setup

1. It is recommended that you review [Lab 1 – Application Guide](#) before starting this lab.
2. Turn on the QBot Platform by pressing the power button once. To ensure the robot is ready for the lab, check the following conditions:
 - a. Confirm that LEDs on the robot base are solid red, and the LCD shows a battery voltage above 12.5 V.
 - b. The Logitech F710 joystick's wireless receiver is connected to the QBot Platform. Ensure the switch on the joystick is in the **X position** and that the LED beside the Mode button is **off**.
 - c. Connect your computer to the same network that the QBot Platform is connected to. If using the provided router, connect to the Wi-Fi network **Quanser_UVS-5G**.
3. Test connectivity: Open a command prompt on your local computer, enter the following command, replacing `192.168.x.x` with the QBot's IP address found on its LCD display, and hit enter: `ping 192.168.x.x`

A successful connection returns a line such as: `Reply from 192.168.x.x: bytes=32 time=20ms TTL=64.`
4. In Section A of `for_kin.py`, set `ipHost` to your Windows computer's IPv4 address. To find it, run `ipconfig` in Command Prompt and look under your active network connection. Your IP address is listed as the IPv4 address, and should look like the one for the QBot, `192.168.X.X`.

Running Code

You will edit the code on your local Windows machine and then transfer it to the QBot Platform. Repeat the steps in this section each time you change the code. **Refer back to this section later when instructed to run the code.**

1. Open WinSCP, use the QBot's IP address as the **Host name**, and "nvidia" for both Username and Password. Select Login. This opens a connection to the QBot for file transfer.
2. In the WinSCP window, transfer the following files from your local Windows machine to the QBot:

- a. On the QBot, open
/home/nvidia/Documents/Quanser/6_teaching/3_Robotics/sp1_task_automation/l2_forward_kinematics/hardware/python, then copy the updated [for_kin.py](#) and [qbot_platform_driver_physical.rt-linux_qbot_platform](#) into that directory.
- b. Navigate to the directory
/home/nvidia/Documents/Quanser/0_libraries/python/ shown in Figure 1, or create these directories if they do not exist. Copy over the “hal” and “pal” folder from your local Windows machine, replacing existing files on QBot.

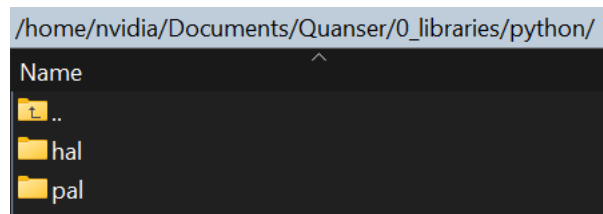


Figure 1: 'hal' and 'pal' folder in correct directory

3. Run [observer.py](#) on your local Windows machine first to initiate receiving data feeds from the QBot Platform. Keep this script running on your Windows machine while you run [for_kin.py](#) in the next step.
4. Run [for_kin.py](#) on QBot Platform:
 - a. Open a PuTTY session: at the top menu bar, click Commands > Open in PuTTY.
 - b. In the PuTTY terminal, enter the password “nvidia”.
 - c. Navigate to the python script directory from step 2a by using the `cd` command.
 - d. Run the script using the following command:

```
sudo PYTHONPATH=$PYTHONPATH python3 for_kin.py
```

The LEDs on the base of the QBot will turn blue when the scripts starts successfully.
 - a. To stop both [for_kin.py](#) and [observer.py](#), press the right button (RB) on the joystick.

Drive with Individual Wheel Speeds Command

1. Run the code following the steps in the [Running Code](#) section. Press and hold the left button (labelled LB) to arm the robot. Notice that the LEDs turn green. Keep this button pressed as you teleoperate the QBot for future labs as well.

Note: If your robot is in a position where it may collide with obstacles or people, disarm

the robot by releasing the LB button. The LEDs will turn blue again, indicating that the QBot Platform is disarmed.

2. While armed, use the following joystick sticks to move wheels while monitoring the **Motor Speed Plot** and **Body Speed Plot** window as shown in Figure 2. Take note of the wheel speeds, as well as what motion it produces for the QBot Platform body. **Body Speed Plot** displays turn speed from the gyroscope and body speeds from the forward kinematics model which you will compete later.
 - a. move only the left stick up and down and determine the positive convention of the left wheel speed.
 - b. move only the right joystick up and down and determine the positive convention of the right wheel speed.
 - c. move both sticks to drive the QBot around. Take note of the stick combination that produce the following results:
 - i. forward or backward motion
 - ii. turn in place clockwise or counterclockwise.

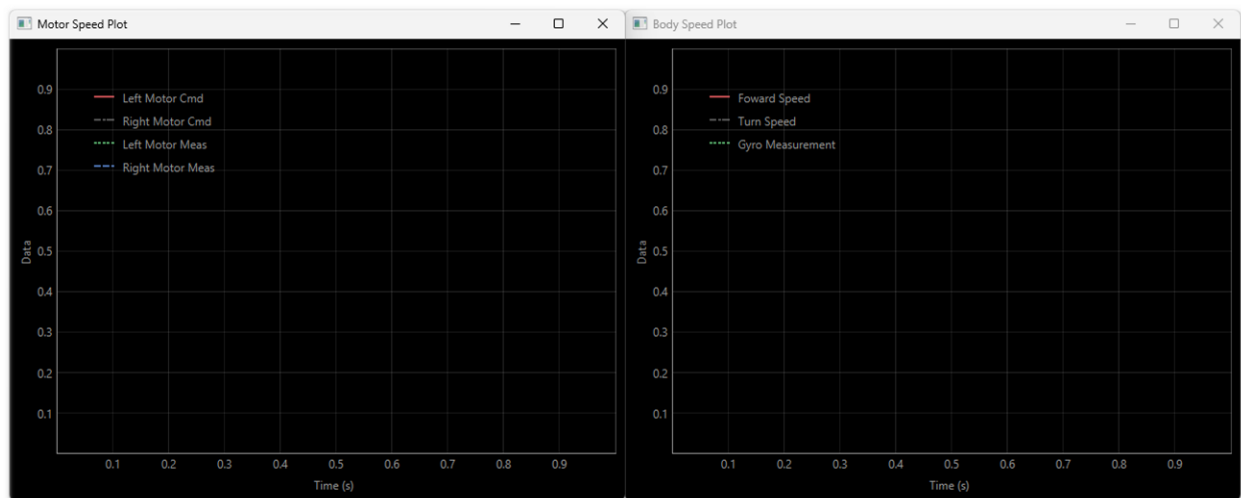


Figure. 2. QBot Speeds Scope

3. What relationship do you observe between the wheel speeds and turn speed of the robot?
4. Stand behind the robot and use both joysticks to control the QBot to follow the lines on the mat (straight or curved, or both). Take notes on the challenged experienced during this activity.
5. Using [observer.py](#), open the downward facing camera feed, shown in Figure 3. Now, instead of walking behind the QBot, try making the robot follow the lines on the mat

using only the downward facing camera feed. Document the strategies you utilized to follow the line and additional challenges faced when using only the camera feed.

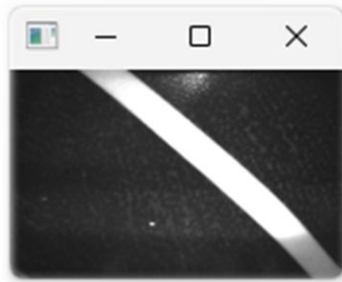


Figure. 3. Downward Facing Camera Feed

6. Stop the code by pressing the right button (RB) on the joystick.

Forward Kinematics Formulation

1. Go back to your local Windows machine and open [for_kin.py](#). In Section B, right click on the **QBPMovement()** class and select "Go to Definition". This class handles the computation of forward kinematics as well as things needed in future labs, including inverse kinematics and line following, but these methods are incomplete. This class also contains physical QBot parameters.
2. Using the predefined global variables with the QBot's physical dimension, complete **diff_drive_forward_velocity_kinematics()** function under **QBPMovement()** to formulate the QBot forward kinematics equation via geometric derivation.
3. Save your changes to **QBPMovement()**. Using the function you completed on the previous step, go back to the [for_kin.py](#) file and complete the missing code in Section F. Note: The 'QBPMovement()' class was instantiated as 'movement' in Section B. This object can be used to reference the functions within the class.
4. Run the code following the steps in the [Running Code](#) section and drive the QBot.
5. Using the scope display window, verify that the turn speeds output from your forward kinematics function closely matches that from the gyroscope. Revise your function if needed. When you're satisfied with the output of your function, take a screenshot of the scope window before shutting down the script. Also monitor the forward speed of the robot. What is the maximum forward or turn speed of the robot?
6. Stop your script using the RB button. Ensure that you save a copy of your completed files for review later.
7. Turn OFF the QBot by single pressing the power button. Do not hold it down. After shutdown, confirm that all the LEDs are off.